

Advanced Machine learning Mastering Course

Introduced by

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Agenda

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2 Introduction to Regression

3 More Regression

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10 Neural Networks

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11 Math behind SVMs

11 SVMs in Practice

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15 Bayesian Inference

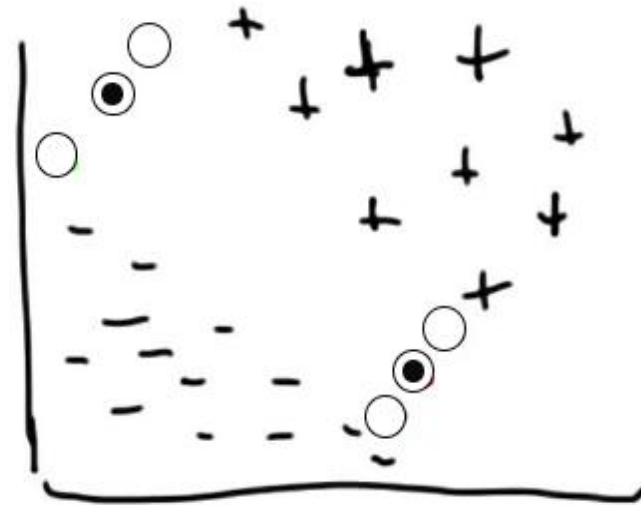
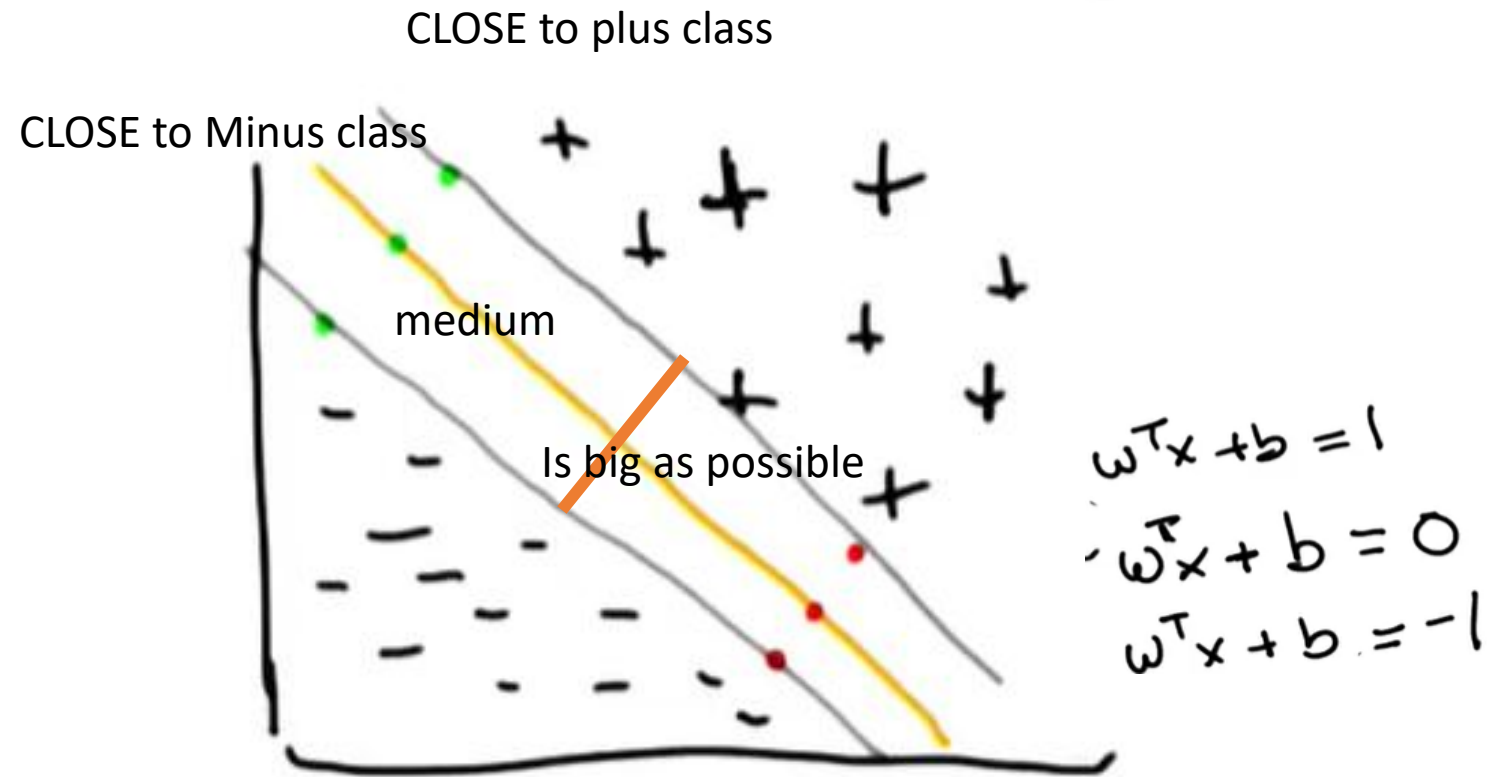
16 Ensemble B&B

17 Finding donors for CharityML

$$\begin{aligned}\omega^T x_1 + b &= 1 \\ \omega^T x_2 + b &= -1\end{aligned}$$

QUIZ!

WHICH IS THE BEST



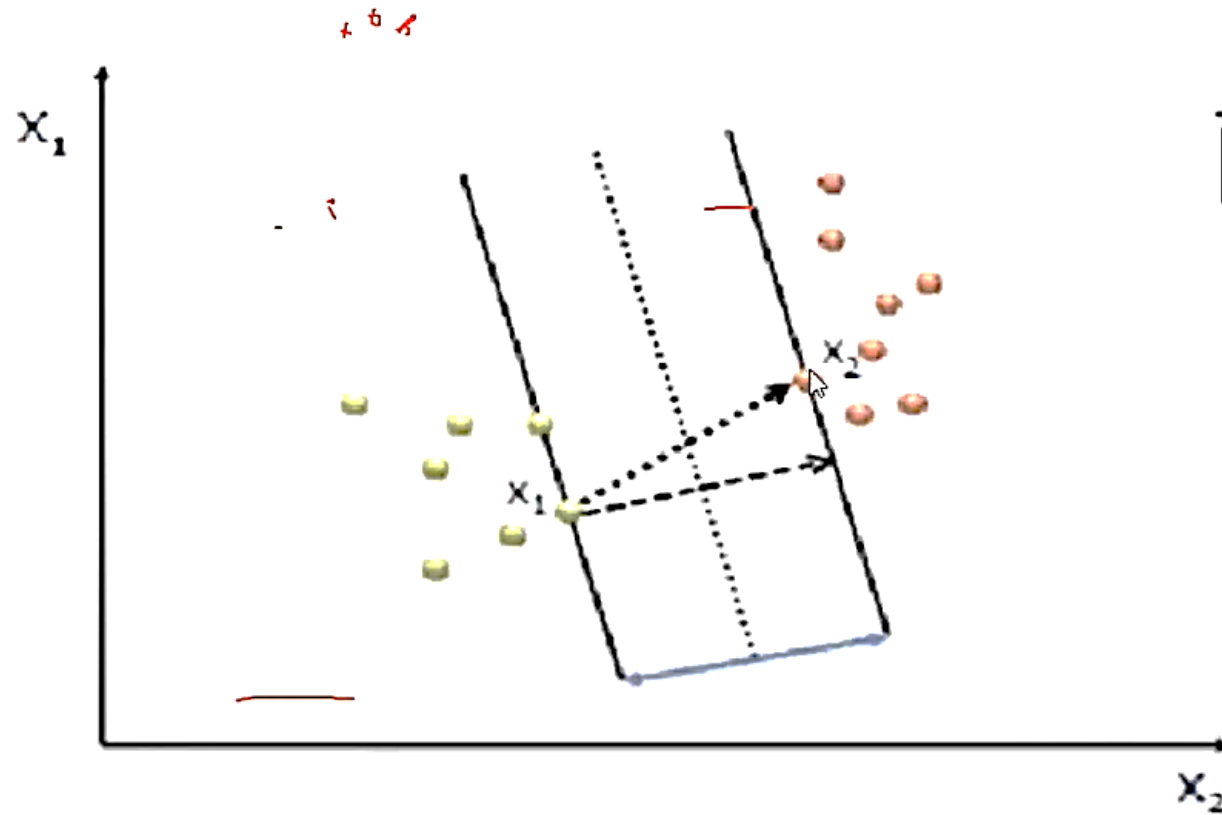
Quiz

$$\begin{array}{r}
 \omega^T x_1 + b = 1 \\
 - \quad \omega^T x_2 + b = -1 \\
 \hline
 \boxed{\omega^T x_1 - \omega^T x_2 = 2}
 \end{array}$$

$$\omega^T(x_1 - x_2) = 2$$

$$\frac{\omega^T(x_1 - x_2)}{\|\omega\|} = \frac{2}{\|\omega\|}$$

$$\underbrace{\frac{\omega^T(x_1 - x_2)}{\|\omega\|}}_{\text{margin}} = \frac{2}{\|\omega\|}$$



$$\frac{w}{\|w\|} \cdot (x_2 - x_1) = \text{width} = \frac{2}{\|w\|}$$

$$w \cdot x_2 + b = 1$$

$$w \cdot x_1 + b = -1$$

$$w \cdot x_2 + b - w \cdot x_1 - b = 1 - (-1)$$

$$w \cdot x_2 - w \cdot x_1 = 2$$

$$\frac{w}{\|w\|} (x_2 - x_1) = \frac{2}{\|w\|}$$

max $\frac{2}{\|w\|}$ while classifying everything correctly
 $y_i (w^T x_i + b) \geq 1 \quad \forall i$

• min $\frac{1}{2} \|w\|^2$

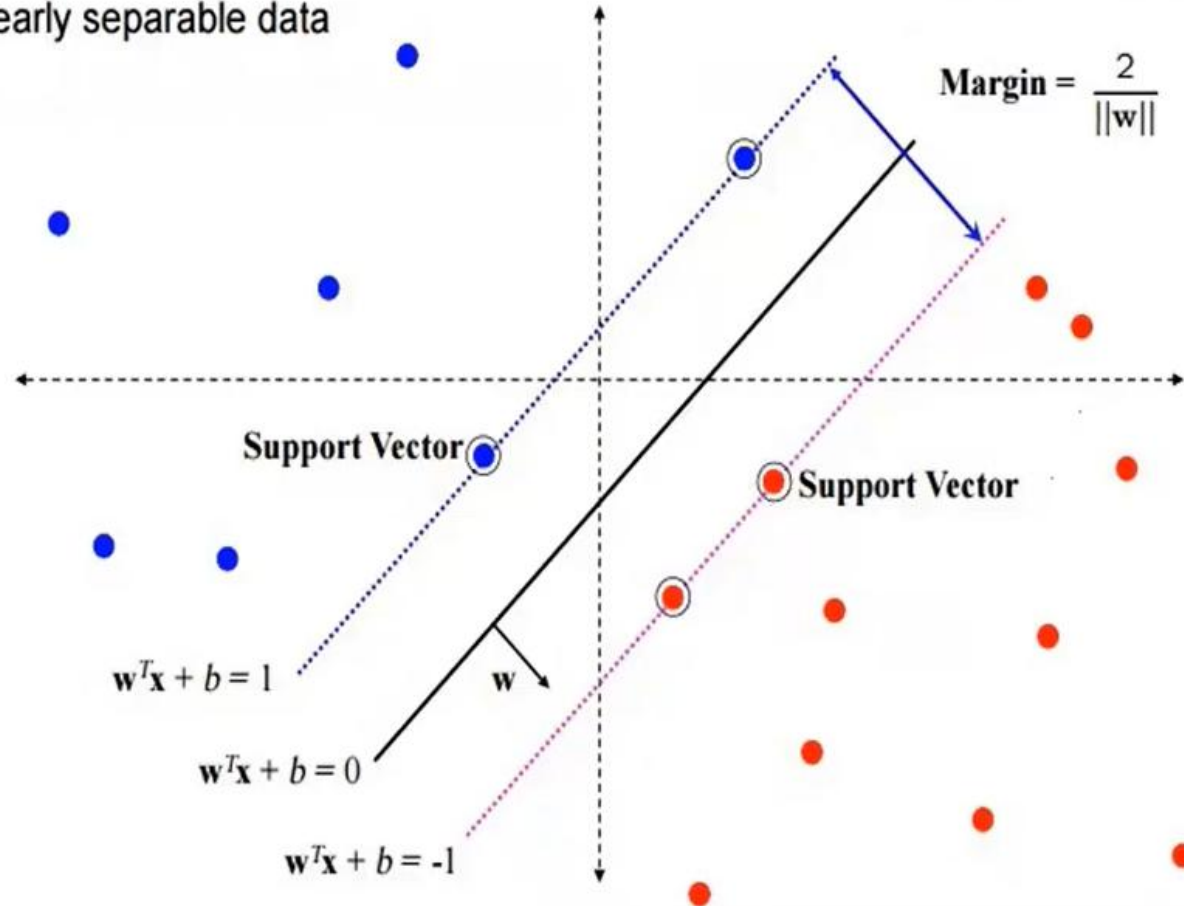
quadratic programming

•
$$W(\alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

max st. $\alpha_i \geq 0, \sum_i \alpha_i y_i = 0$

Support Vector Machine

linearly separable data



Support Vector Machines (SVM) can be framed as a minimization problem in the context of both classification and regression. The objective is to find the optimal hyperplane that separates data points of different classes with the maximum margin in the case of classification, or to fit the best possible regression line with minimal error in the case of regression. Here's how SVM can be formulated as a minimization problem for both cases:

1. SVM for Classification

Primal Form

The primal form of the SVM optimization problem can be written as:

Minimize:

$$\frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i$$

Subject to:

$$y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \forall i$$

$$\xi_i \geq 0, \quad \forall i$$

where:

- $\mathbf{w}$ is the weight vector.
- b is the bias term.
- C is the regularization parameter that controls the trade-off between maximizing the margin and minimizing the classification error.
- ξ_i are the slack variables that allow for some misclassification.
- y_i are the class labels (+1 or -1).
- $\mathbf{x}_i$ are the feature vectors of the training samples.

Dual Form

The dual form of the SVM optimization problem can be written as:

Maximize:

$$\sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \mathbf{x}_i \cdot \mathbf{x}_j$$

Subject to:

$$\sum_{i=1}^n \alpha_i y_i = 0$$

$$0 \leq \alpha_i \leq C, \quad \forall i$$

where:

- α_i are the Lagrange multipliers.

$$\min \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{x}_i \cdot \mathbf{w} + b) - 1 \geq 0 \quad \forall_i$$

In order to cater for the constraints in this minimization, we need to allocate them Lagrange multipliers α , where $\alpha_i \geq 0 \quad \forall_i$:

$$\begin{aligned} L_P &\equiv \frac{1}{2} \|\mathbf{w}\|^2 - \alpha [y_i(\mathbf{x}_i \cdot \mathbf{w} + b) - 1 \quad \forall_i] \\ &\equiv \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^L \alpha_i [y_i(\mathbf{x}_i \cdot \mathbf{w} + b) - 1] \\ &\equiv \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^L \alpha_i y_i (\mathbf{x}_i \cdot \mathbf{w} + b) + \sum_{i=1}^L \alpha_i \end{aligned}$$

We wish to find the $\mathbf{w}$ and b which minimizes, and the α which maximizes L_P (whilst keeping $\alpha_i \geq 0 \quad \forall_i$). We can do this by differentiating L_P with respect to $\mathbf{w}$ and b and setting the derivatives to zero:

$$\frac{\partial L_P}{\partial \mathbf{w}} = 0 \Rightarrow \mathbf{w} = \sum_{i=1}^L \alpha_i y_i \mathbf{x}_i$$

$$\frac{\partial L_P}{\partial b} = 0 \Rightarrow \sum_{i=1}^L \alpha_i y_i = 0$$

2. SVM for Regression (Support Vector Regression, SVR)

In the case of SVR, the objective is to find a function $f(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} + b$ that deviates from the actual observed values y_i by at most ϵ for all training data points.

Primal Form

The primal form of the SVR optimization problem can be written as:

Minimize:

$$\frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*)$$

Subject to:

$$y_i - (\mathbf{w} \cdot \mathbf{x}_i + b) \leq \epsilon + \xi_i$$

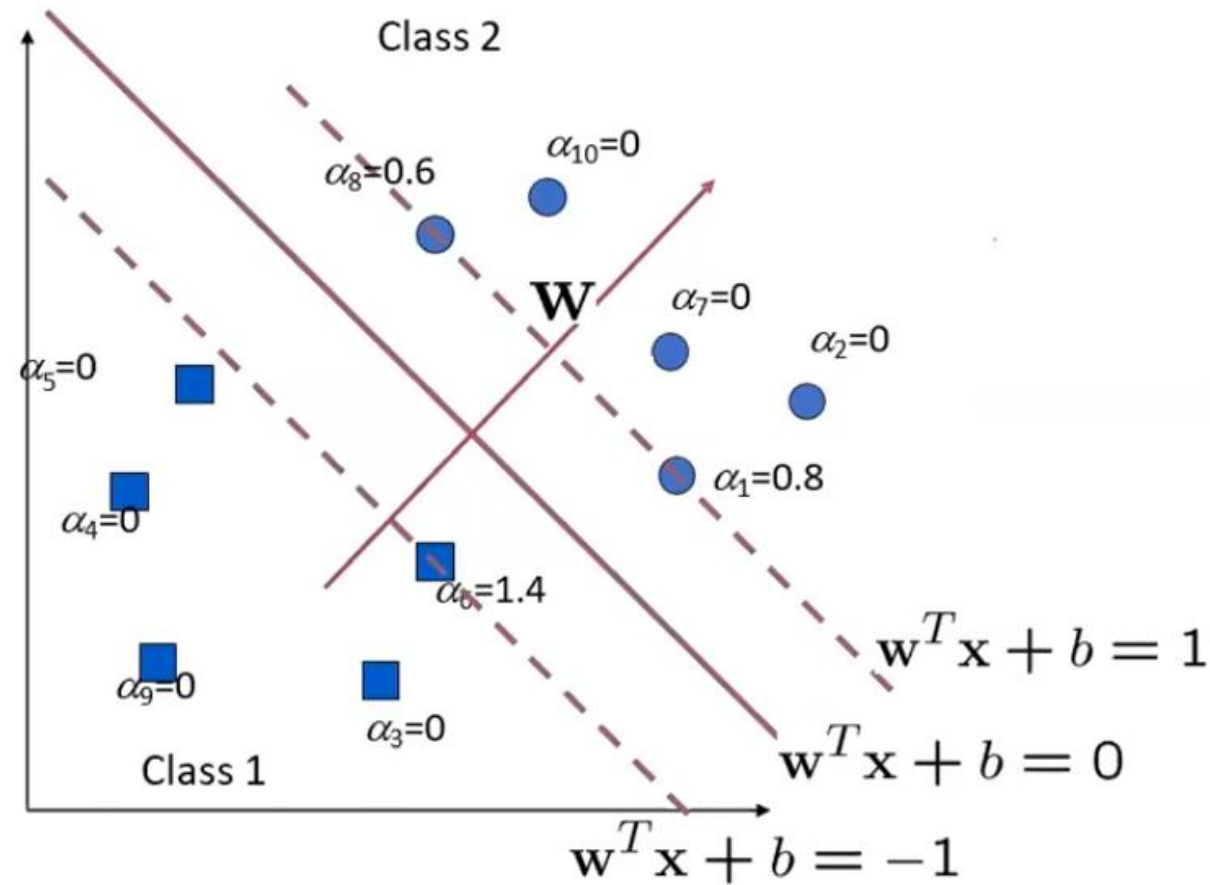
$$(\mathbf{w} \cdot \mathbf{x}_i + b) - y_i \leq \epsilon + \xi_i^*$$

$$\xi_i, \xi_i^* \geq 0, \quad \forall i$$

where:

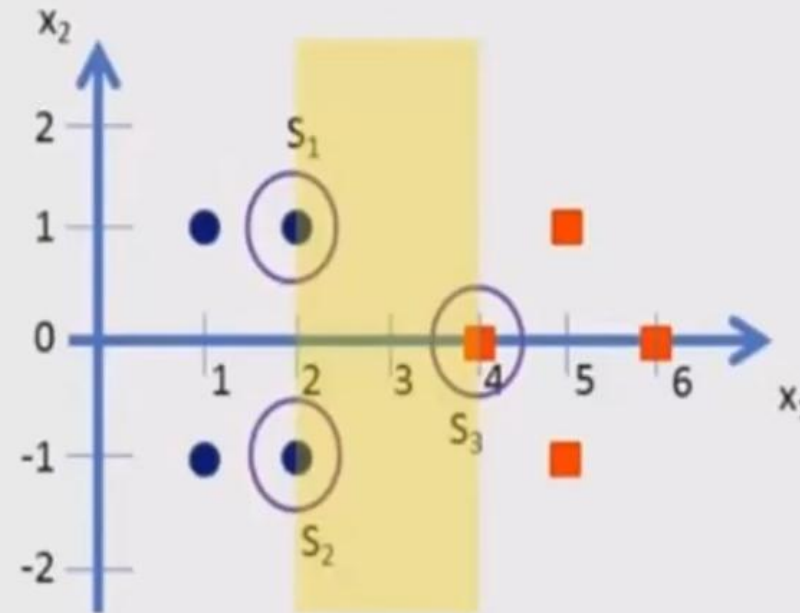
- $\mathbf{w}$ is the weight vector.
- b is the bias term.
- C is the regularization parameter.
- ξ_i and ξ_i^* are the slack variables that measure the deviations of predictions from the actual values.
- ϵ is the margin of tolerance.

A Geometrical Interpretation



Example

- Here we select 3 Support Vectors to start with.
- They are S_1 , S_2 and S_3 .



$$S_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$S_2 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$S_3 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

$$S_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$S_2 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$S_3 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

$$\widetilde{S}_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

$$\widetilde{S}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$$\widetilde{S}_3 = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$$

Now we need to find 3 parameters α_1 , α_2 and α_3 based on the following 3 linear equations:

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_1 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_1 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_1 = -1 \text{ (-ve class)}$$

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_2 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_2 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_2 = -1 \text{ (-ve class)}$$

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_3 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_3 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_3 = +1 \text{ (+ve class)}$$

Let's substitute the values for $\tilde{S}_1$, $\tilde{S}_2$ and $\tilde{S}_3$ in the above equations.

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} = +1$$

After simplification we get

$$6\alpha_1 + 4\alpha_2 + 9\alpha_3 = -1$$

$$4\alpha_1 + 6\alpha_2 + 9\alpha_3 = -1$$

$$9\alpha_1 + 9\alpha_2 + 17\alpha_3 = +1$$

Simplifying the above 3 simultaneous equation we get

$$\alpha_1 = -3.25 \quad \alpha_2 = -3.25 \quad \text{and} \quad \alpha_3 = 3.5$$

The hyperplane that discriminates the positive class from negative class is given by:

$$\tilde{w} = \sum_i \alpha_i \tilde{S}_i$$

$$\tilde{w} = \alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$$

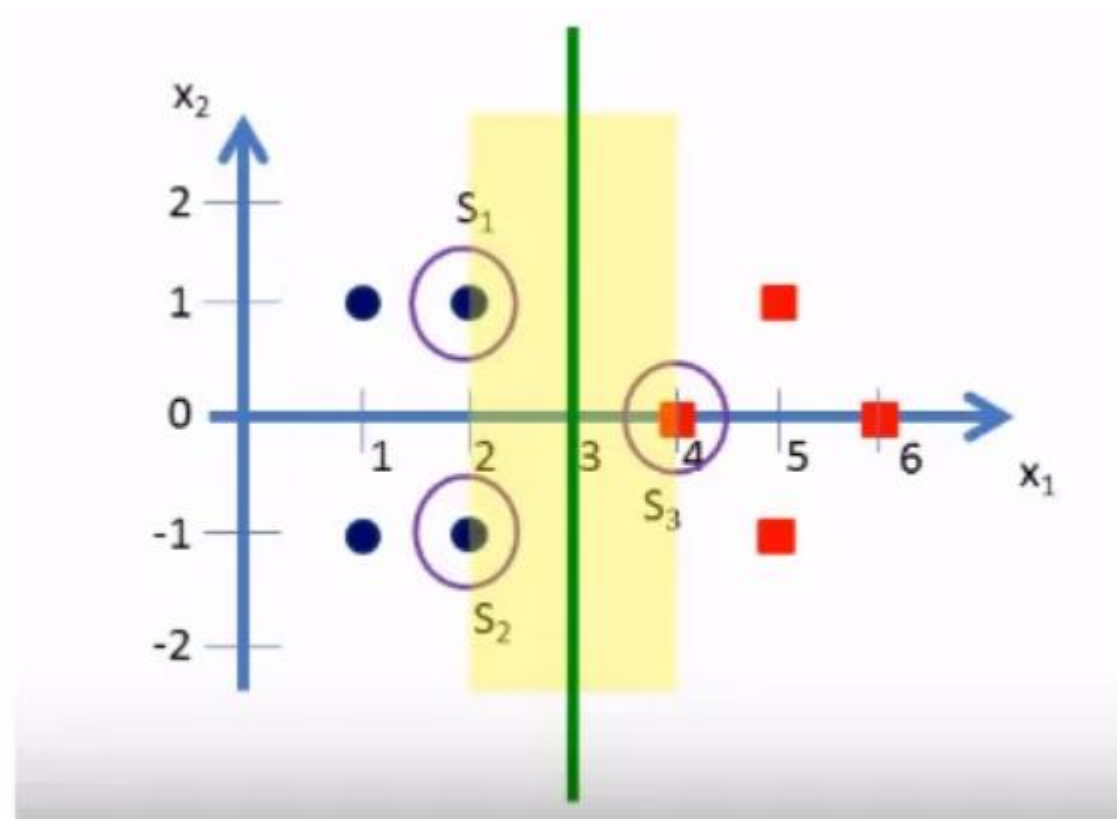
$$\tilde{w} = (-3.25) \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + (-3.25) \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + (3.5) \cdot \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$$

Our Vectors are augmented with a bias.

Hence we can equate the entry in $\tilde{w}$ as the hyperplane with an offset b .

Therefore the separating hyperplane equation

$$\mathbf{y} = \mathbf{w}\mathbf{x} + \mathbf{b} \quad \mathbf{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and offset } \mathbf{b} = -3$$

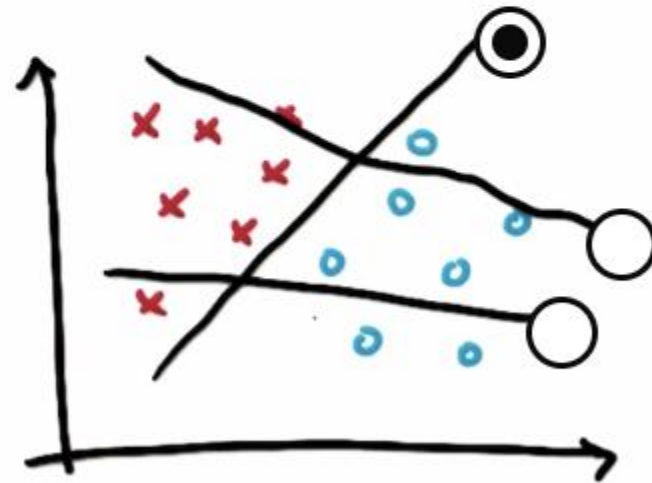


This is the expected decision surface of the LSVM.

Quiz

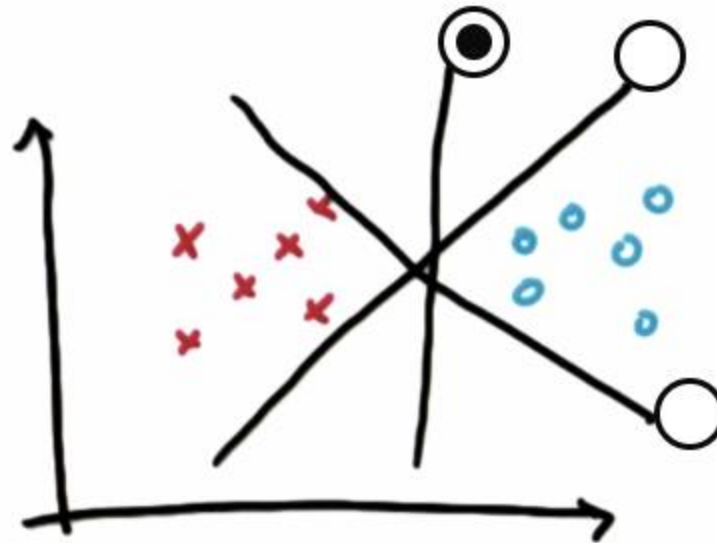
Separating Line

SUPPORT VECTOR MACHINE



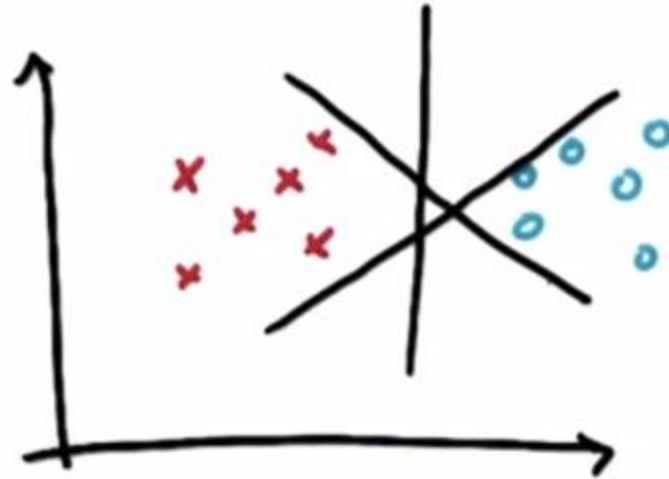
Choosing Between Separating Lines

SUPPORT VECTOR MACHINE



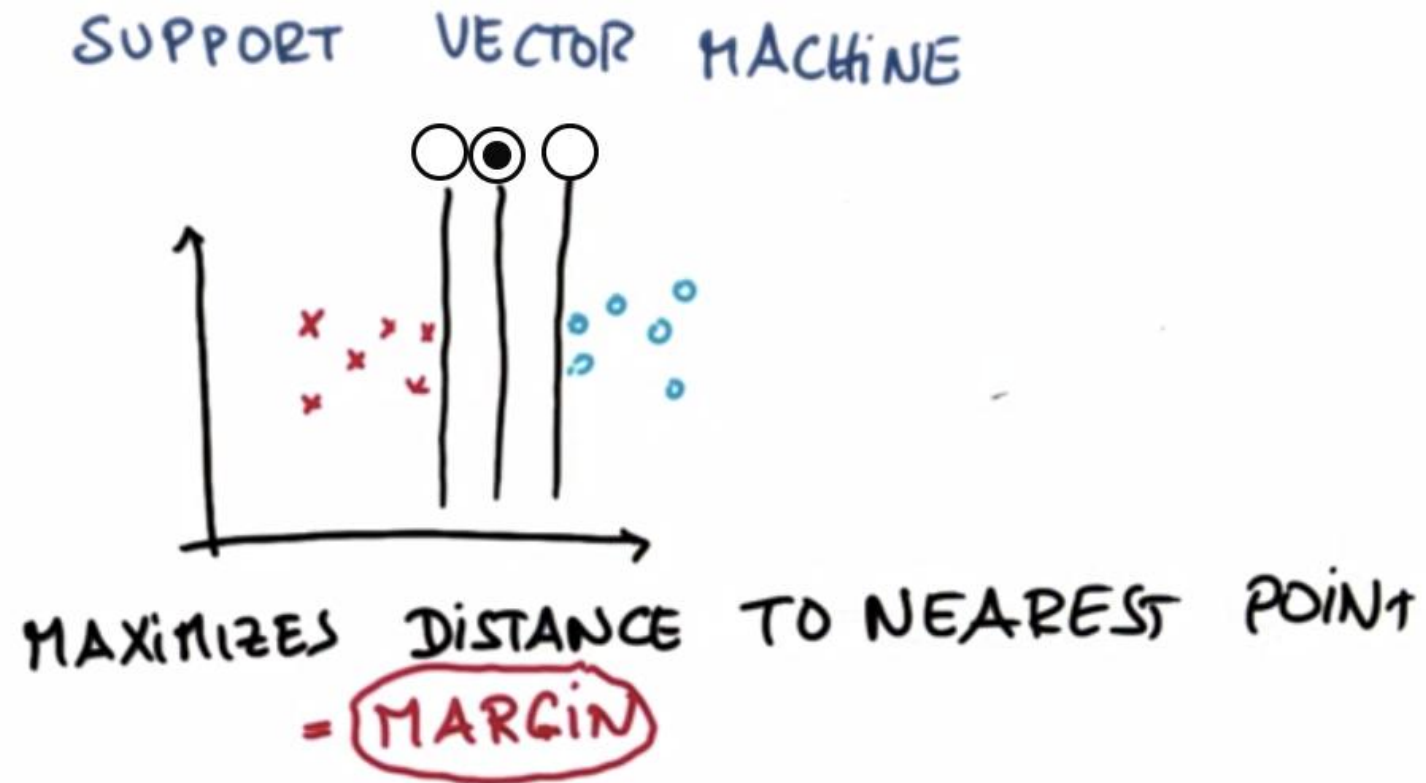
What Makes A Good Separating Line

SUPPORT VECTOR MACHINE



- ☐ SIMPLE
- ☐ RANDOM
- ☒ SOMETHING ELSE

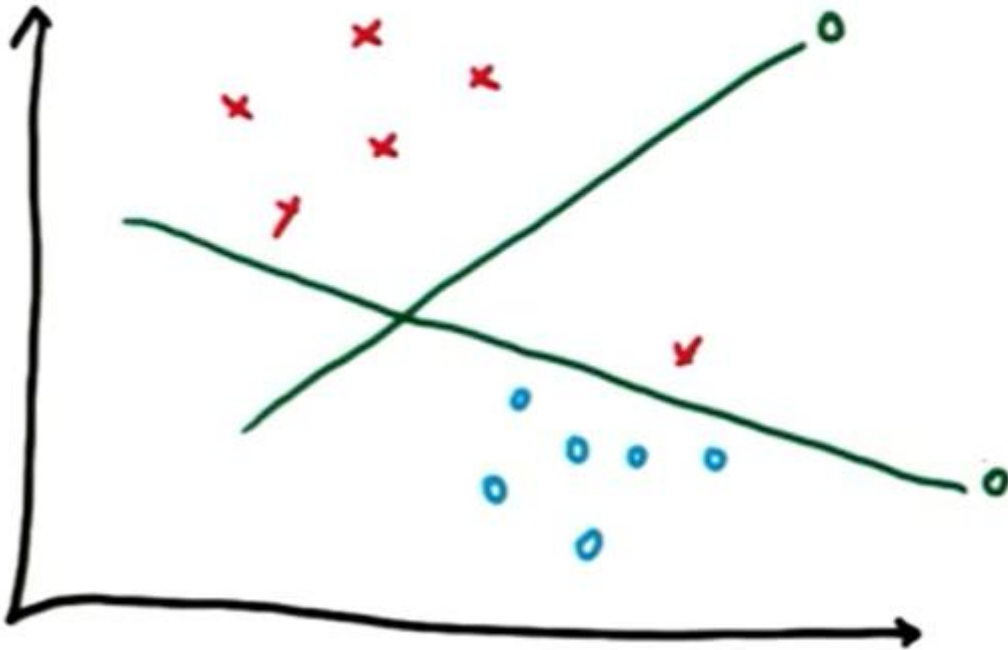
Practice with Margins



Which is the best line

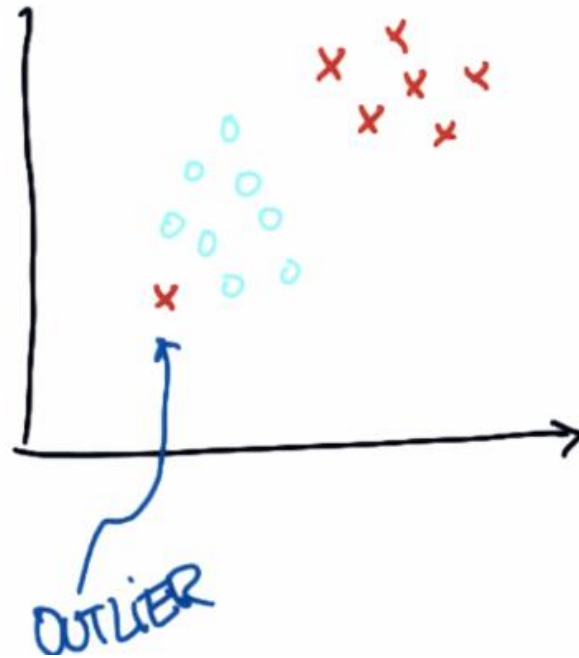
SUPPORT VECTOR MACHINES

svm puts first and foremost the correct classification of the labels and then maximize the margin
so for svm you are trying to classify correctly and subject to that constrain you maximize the margin



SVM Response to Outliers

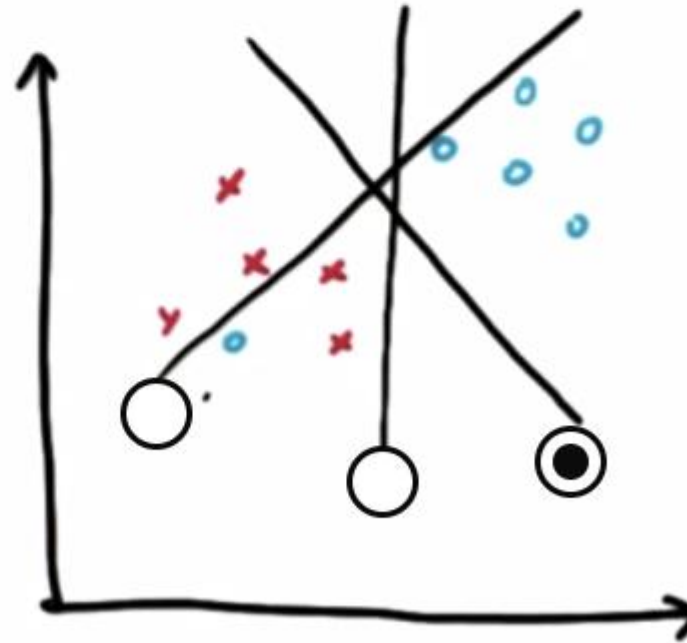
SVMs - OUTLIERS



- GIVE UP
- SAY SOMETHING'S RANDOM
- DO THE BEST IT CAN

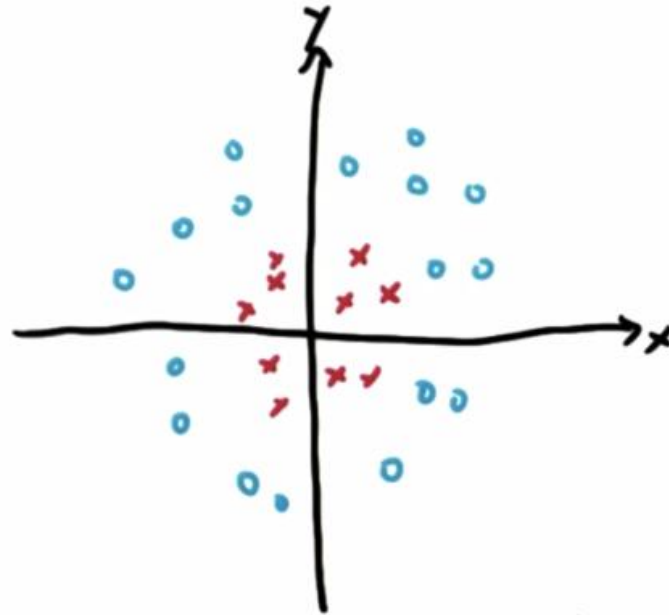
SVM Outlier Practice

SVMs - OUTLIERS



Nonlinear Data

SVM

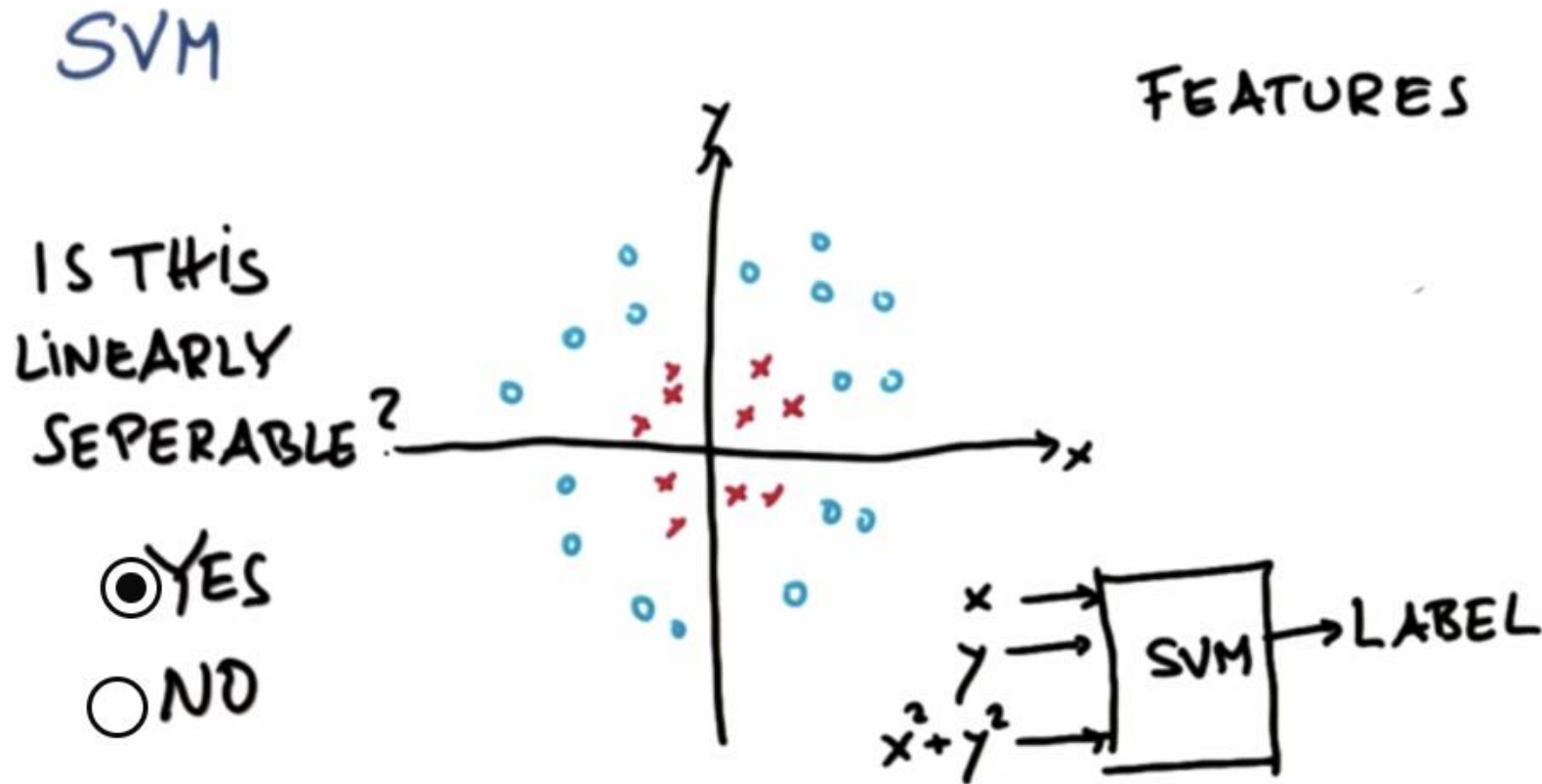


WILL SVMs WORK?

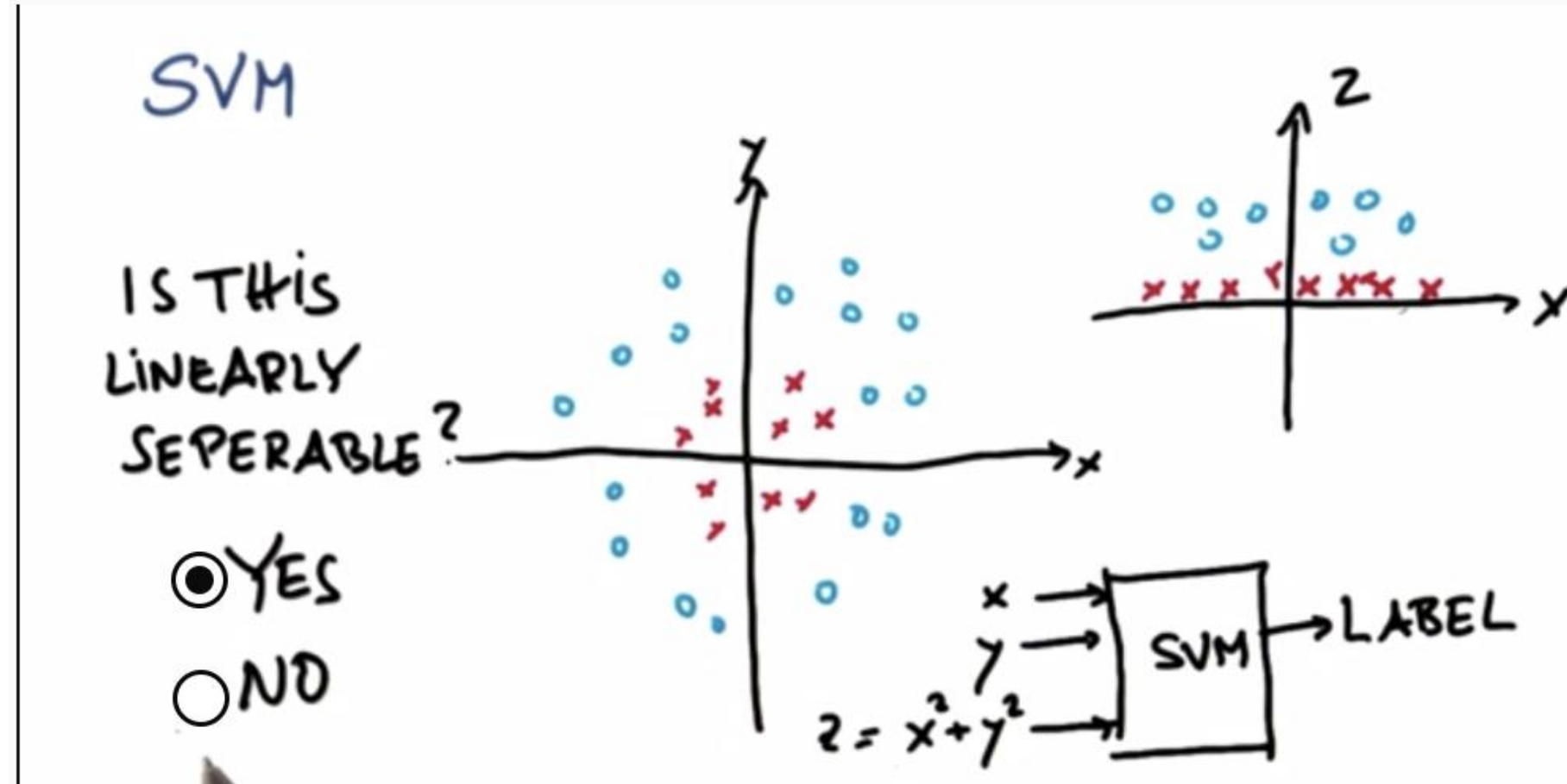
☐ YES

☒ NO

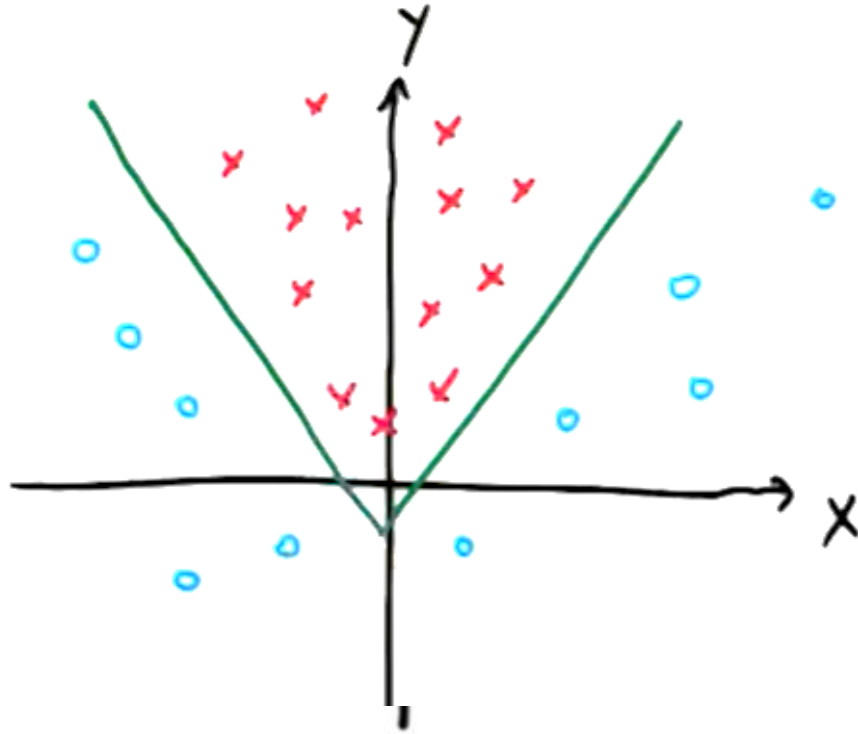
A New Feature



Separating with the New Feature



Practice Making a New Feature



ADD FEATURE^E

o $x^2 + y^2$

o $|x|$

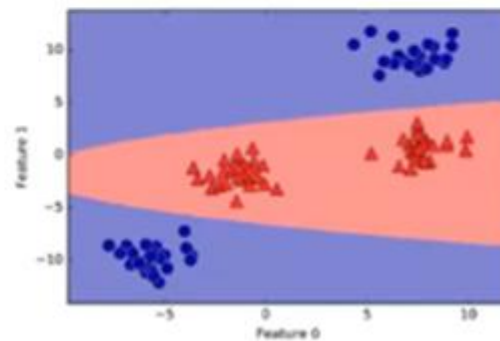
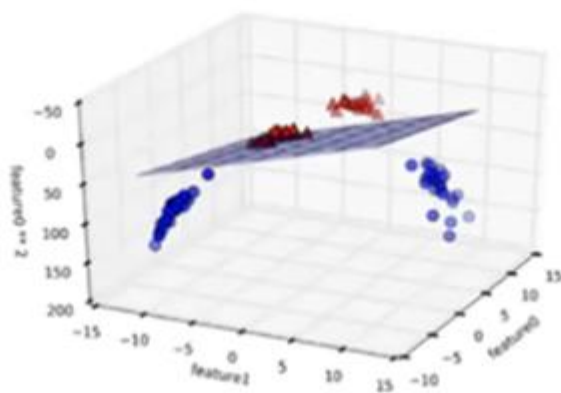
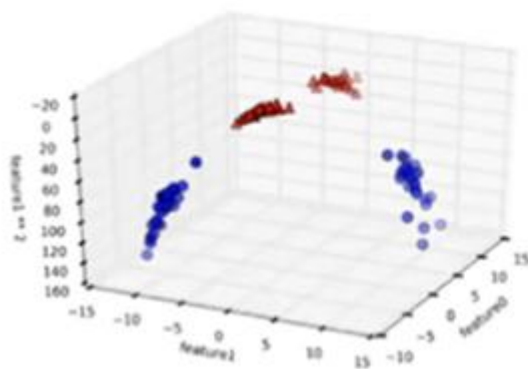
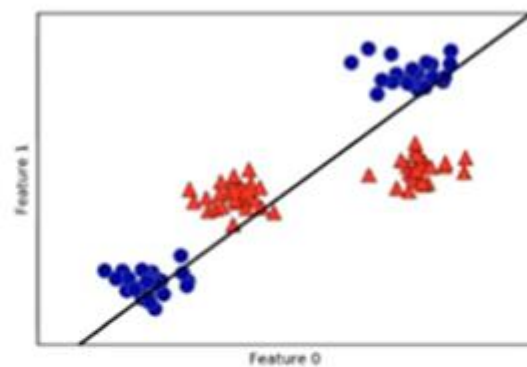
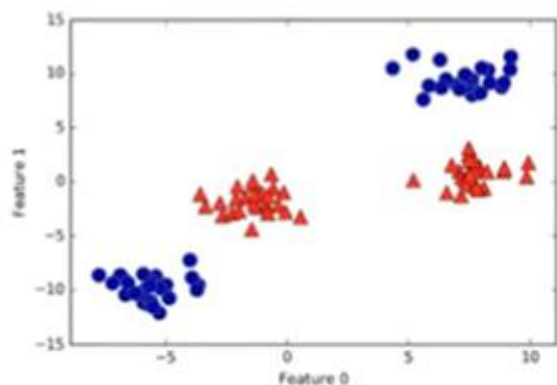
o $|y|$

Overfitting



- ☐ C
- ☐ γ
- ☐ KERNEL

(Applying Kernel Function)



Thank You